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(iii) From expression (1.33), it is clear that the potential due to positive charge decreases as the distance increases, but for a negative charge the potential increases as the distance is increased. At infinity $(r = \infty)$ electrostatic potential is zero (V = 0).

In the case of gravitational force, mass moves from a point of higher gravitational potential to a point of lower gravitational potential. Similarly a positive charge moves from a point of higher electrostatic potential to lower electrostatic potential. However a negative charge moves from lower electrostatic potential to higher electrostatic potential. This comparison is shown in Figure 1.24.

(iv) The electric potential at a point P due to a collection of charges $q_1, q_2, q_3 \ldots . q_n$ is equal to sum of the electric potentials due to individual charges.

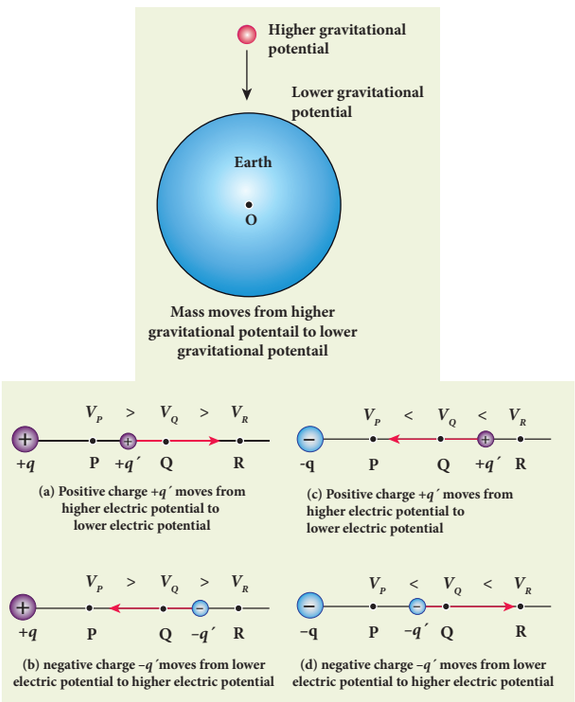


Figure 1.24 Motion of charges in terms of electric potential

$$V_{tot} = \frac{kq_1}{r_1} + \frac{kq_2}{r_2} + \frac{kq_3}{r_3} + \ldots$$
$$\ldots . + \frac{kq_n}{r_n} = \frac{1}{4\pi\varepsilon_0} \sum_{i=1}^{n} \frac{q_i}{r_i} \qquad (1.34)$$

where $r_1, r_2, r_3 \ldots\ldots r_n$ are the distances of q_1, q_2, q_3 ….. q_n respectively from P (Figure 1.25).

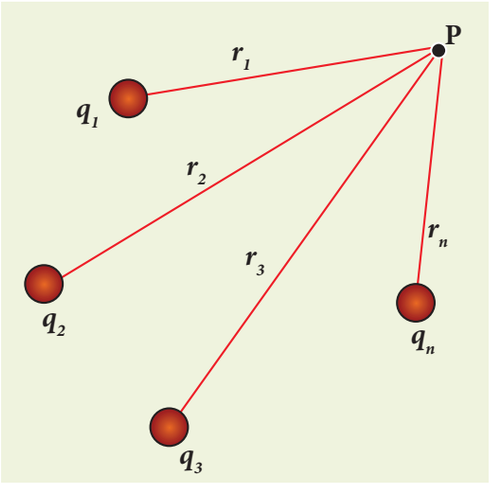
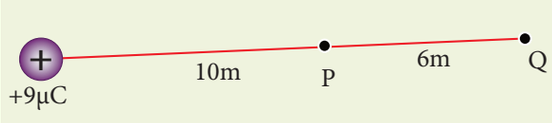


Figure 1.25 Electrostatic potential due to collection of charges

EXAMPLE 1.12

(a) Calculate the electric potential at points P and Q as shown in the figure below.

(b) Suppose the charge +9μC is replaced by -9μC find the electrostatic potentials at points P and Q



(c) Calculate the work done to bring a test charge +2μC from infinity to the point P. Assume the charge +9μC is held fixed at origin and +2μC is brought from infinity to P.

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